

Pilot Study Report

Regime-Switching Risk Models for ETF Portfolios — Pilot Prepared for Prof. Seo. All numbers below are read programmatically from `output/tables/*.json`; see `scripts/generate_pilot_report.py`.

1. Question and why a pilot

For a fixed equal-weight SPY/IEF/GLD portfolio, does letting cross-asset *correlations* change by regime improve one-day-ahead VaR/Expected Shortfall forecasts beyond letting only *volatilities* change? A pilot was run first because the correlation channel is the harder, more novel half of the design (Section 6 of `docs/literature_briefing.md` shows the published evidence on this is genuinely split), and because the full 5-week thesis timeline does not survive discovering a feasibility, power, or reproducibility problem in week 3 instead of week 1.

2. What was run

Five work packages: WP0 (data acquisition and audit of SPY/IEF/GLD, 2010-2025), WP1 (a 41-test correctness suite for the model engine), WP2 (a Monte Carlo power study — can this design detect a correlation-regime effect at all, given ~10 years of daily data?), WP3 (a real-data pilot on the locked-safe 2010-2015 window: descriptive regime statistics, walk-forward backtest, stability diagnostics, named-episode diagnostics), and WP4 (a single-machine reproducibility check).

Component	Setting
Universe (pilot)	SPY, IEF, GLD, equal-weight, no rebalancing
Regime model	2-state Gaussian HMM on SPY returns only, filtered/one-step-predictive probabilities
Covariance variants	A (no regime), B (regime vol, pooled corr), C (pooled vol, regime corr), D (regime vol + corr)
Benchmarks	Historical simulation (500d), EWMA (lambda=0.94), multivariate EWMA, GARCH(1,1)-t
Estimation window / refit	1260 trading days, refit every 21 days (GARCH every 63)
Alpha levels	5% (primary), 2.5% (secondary)
Scoring	FZ0 joint VaR/ES loss, Diebold-Mariano with Newey-West HAC errors
WP2 simulation scenarios	S0 vol-only (null), S1 moderate breakdown, S2 strong breakdown, S3 flight-to-quality; stylised + real-data-calibrated DGP; 200 reps @5%, 100 reps @2.5% each

3. Results

3a. Correctness (WP1)

41 passed, 80 warnings in 1.86s — all tests pass, covering the mixture VaR/ES root-finding, the zero-mean fix, HMM warm-starting, the no-look-ahead invariant, and the EWMA/EWMA_MV algebraic-equivalence property (see `docs/decisions_log.md`).

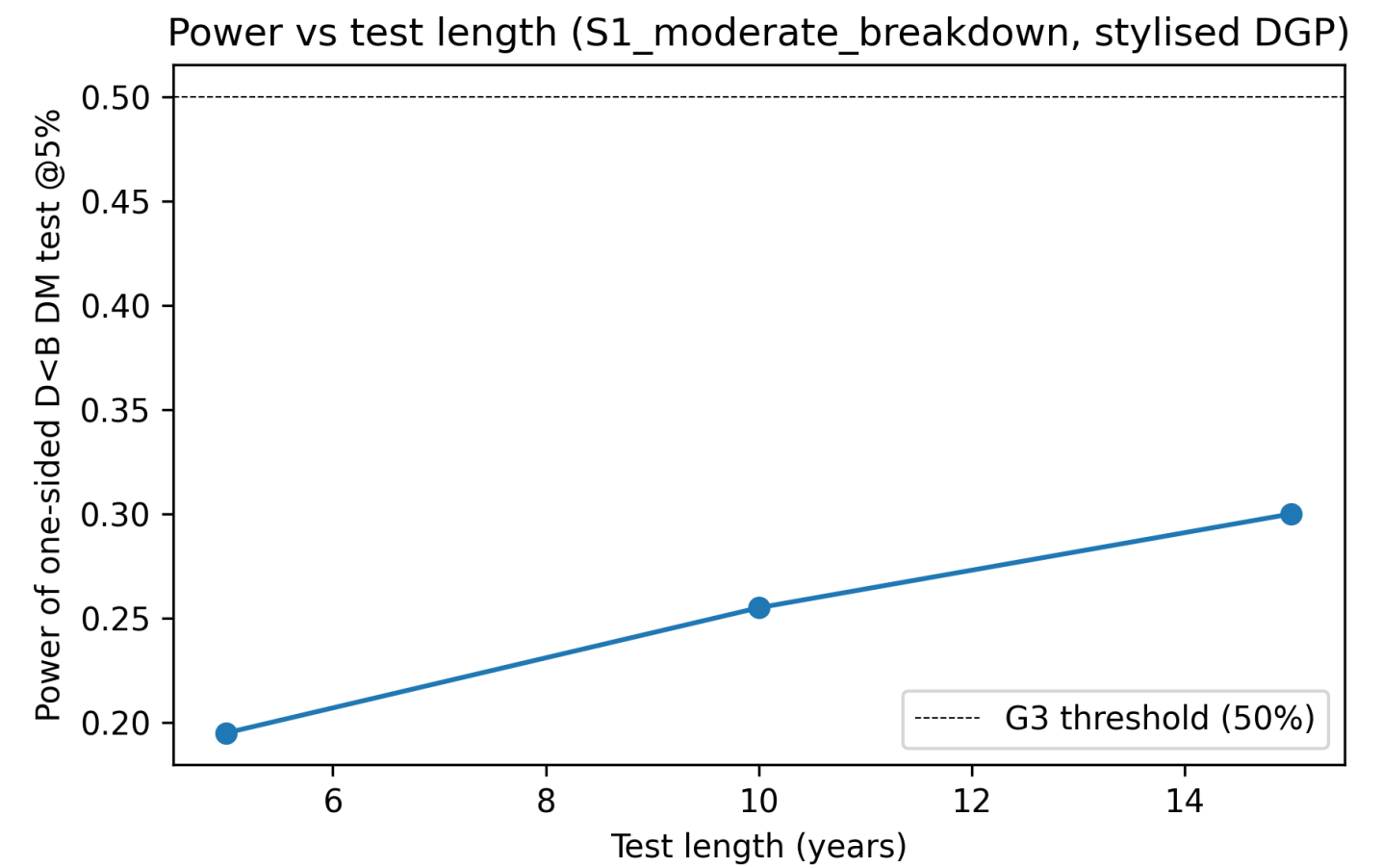
3b. Simulation power (WP2)

D<B one-sided Diebold-Mariano test, rejection rate at nominal alpha=0.05, n=200 replications/cell:

Scenario	Stylised DGP power	Calibrated DGP power
S0 vol-only (null / false-positive rate)	5.0%	2.0%
S1 moderate breakdown	25.5%	16.5%
S2 strong breakdown	52.5%	42.0%
S3 flight-to-quality	3.5%	9.5%

Power vs. test-length (stylised DGP, S1 moderate breakdown, alpha=0.05):

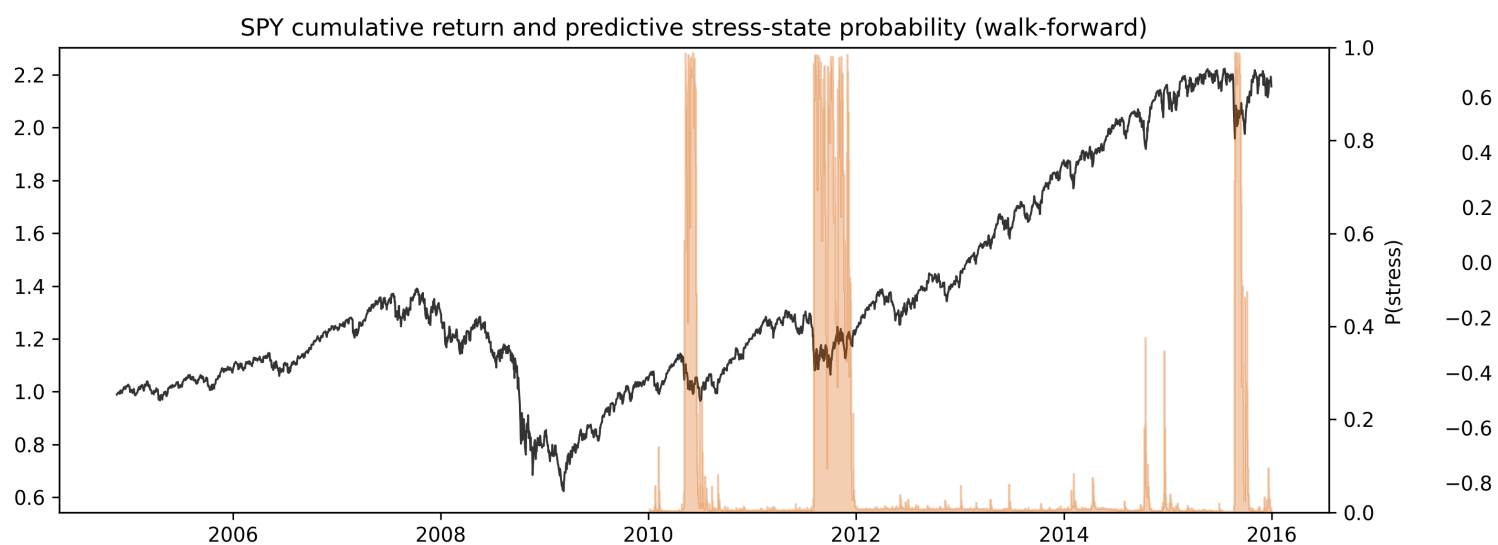
Years of daily data	5	10	15
D<B power	19.5%	25.5%	30.0%



Total WP2 runtime: 40.3 minutes (2417s), 2,800 replications, on this machine (see Section 6 for machine specs).

3c. Real-data pilot, 2010-2015 (WP3)

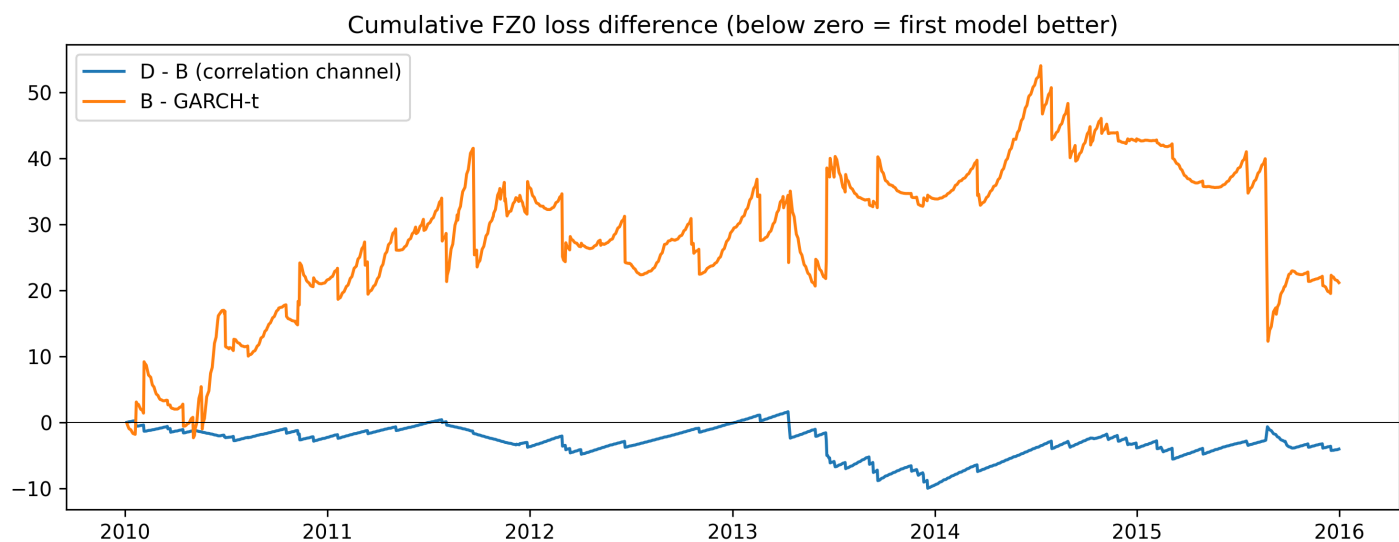
Descriptive: full-sample (2004-2015) stress-regime share 18.3%, expected calm duration 144.7 days, expected stress duration 32.4 days. Shapley attribution of the calm-to-stress portfolio-vol increase: volatility channel 113.1%, correlation channel -13.1% (negative — see Section 5). Equity-bond correlation moves from -0.33 (calm) to -0.51 (stress): *more* negative, i.e. more hedging, not a breakdown, in this sample.



Walk-forward backtest, alpha=5%:

Model	FZ0 loss	Hit rate	Kupiec p	Christoffersen p
A	-4.4058	3.05%	0.0002	0.0133
B	-4.4397	4.24%	0.1637	0.4439
C	-4.3840	2.65%	0.0000	0.0036
D	-4.4423	3.91%	0.0431	0.6488
HS	-4.4129	4.11%	0.1004	0.0148
EWMA	-4.4245	5.10%	0.8599	0.1369
EWMA_MV	-4.4245	5.10%	0.8599	0.1369
GARCHt	-4.4537	5.89%	0.1206	0.1100

Primary contrast D-B: DM $t=-0.469$, $p=0.639$ — not significant on this window. B-GARCHt: $t=0.436$, $p=0.663$; D-GARCHt: $t=0.358$, $p=0.720$ — neither regime variant significantly beats GARCH-t on this window either.



3d. Stability (WP3)

72 monthly refits over 2010-2015. 0/72 collapsed refits, 0/72 large stress-share jumps between refits (no label flips). Expected stress duration per refit ranges 44.6-96.8 days (floor: 5 days). Stress share in-window: mean 17.6%, range [7.7%, 24.2%]. Joint criterion (duration $\geq 5d$ AND stress share in [10%,40%] AND no flip): met in 65/72 refits (90.3%).

3e. Reproducibility (WP4)

WP1 rerun: all passed = `True`. WP3 rerun: numerically identical to committed tables = `True`. WP2 checkpoint-subset recomputation: 0 mismatches (bit-identical, $rtol\ 1e-8$). WP2 under a different master seed (reduced batch, $n=50$ /scenario): same qualitative ranking across scenarios, within the pre-declared wide tolerance for all 4 scenarios (main vs. alt-seed power: S0_vol_only: 5.0% vs 4.0%, S1_moderate_breakdown: 25.5% vs 22.0%, S2_strong_breakdown: 52.5% vs 52.0%, S3_flight_to_quality: 3.5% vs 0.0%). Single-machine scope only (see docs/decisions_log.md).

4. Go/no-go scorecard

Criterion	Verdict	Evidence
G1 Pipeline correctness	PASS	41 passed, 80 warnings in 1.86s
G2 Test size (S0 false-positive $\leq 10\%$)	PASS	Stylised 5.0%, calibrated 2.0%
G3 Detectability (D=50% in S1 or S2, 10y)	PASS (primary DGP)	Stylised S2 52.5%; calibrated DGP does not clear 50% (S2 42.0%)

G4 Regime stability (duration $\geq$ 5d, share 10-40%, $\leq$ 10% failure)	PASS (marginal)	90.3% of refits meet all three jointly
G5 Reproducibility	PASS	WP1 rerun True, WP3 identical True, WP2 subset 0 mismatches
G6 Feasibility (full run $<$ 2h)	PASS	Measured pipeline $\sim$ 26s over a 6-year window; extrapolated $\sim$ 43s for the 10-year 2016-2025 window, even with a 50x safety margin

All six criteria pass -> PROCEED with the full design (Option B). G3 and G4 pass only marginally (G3 fails to clear 50% under the secondary calibrated DGP; G4 clears its 90% bar by 0.3 points) and are reported honestly rather than adjusted after the fact, per the pre-registered rule.

5. What this means

The design is worth running at full scale, but with the power caveat made explicit rather than assumed away. The honest minimum-detectable-effect statement: with 10 years of daily data, this design can reliably ($\geq$ 50% power) detect only a *strong* correlation-regime shift — of the magnitude of the S2 scenario, where calm equity-bond-gold correlations of roughly (-0.30, 0.05, 0.30) move to roughly (0.40, 0.40, 0.30), a full sign flip plus a large magnitude change. A moderate shift (S1: calm correlations moving only to roughly (0.00, 0.20, 0.30)) is detected only 25.5% of the time at 10 years, rising slowly to 30.0% even at 15 years — more data helps, but not quickly enough to rescue a moderate effect within a realistic historical sample. The 2010-2015 real-data window itself turned out to contain no correlation breakdown at all (Shapley correlation share -13.1%, i.e. slightly diversifying), which is a very plausible reason the real-data D-B test comes back null — not evidence the design is broken, but a reminder that whether a correlation-breakdown regime is even present in a given historical window is itself an empirical question, not a given. The full thesis should report power alongside any null the same way WP2 does here.

6. Limitations of the pilot

- **Simulation assumptions:** the DGP is a 2-state Markov-switching multivariate Student-t (df=6) with a fixed, hand-set (or real-data-fitted) transition matrix and stress-correlation target. Real markets may not switch this cleanly between two states, and the true persistence/severity of stress regimes may differ from either DGP calibration used here.
- **Short validation window:** WP3's real-data test is 2010-2015 only (the locked-safe pre-registration window), which happens to contain no strong correlation breakdown. The 2022 stock-bond correlation reversal — the episode motivating this thesis — falls entirely outside this window and is untested by the pilot.
- **Gaussian mixture vs. fat tails:** the VaR/ES engine's regime-conditional covariance variants (A-D) assume a 2-component Gaussian mixture, while the WP2 DGP (and real returns) are fat-tailed (Student-t, df=6). This is a deliberate mismatch worth flagging: any power/loss numbers here are for a mixture-normal forecaster evaluated against fatter-tailed data, not a fully matched specification.
- **Equity-only regime signal:** the HMM is fitted on SPY returns only. The 2013 taper-tantrum diagnostic episode (output/tables/wp3_episodes.json) shows near-zero equity-regime stress probability during a real bond/rates-driven shock — a concrete, real instance of an equity-based signal missing a non-equity-driven regime.
- **EWMA/EWMA_MV algebraic equivalence** (from docs/decisions_log.md): for this fixed-weight portfolio, univariate and multivariate EWMA variance recursions are algebraically identical, so the B/D vs EWMA_MV row in the walk-forward table adds no information beyond B/D vs EWMA. This is a design limitation of using a *fixed*-weight portfolio to test a *multivariate* smooth-correlation benchmark, not a bug, and is reported plainly rather than treated as a redundant confirmation.

7. Next steps: 4-week plan

Week	Deliverable	Decisions needed from Prof. Seo
1	Finalize ETF universe and portfolio construction; extend data to the full 2016-2025 window; re-run WP0-WP1 equivalents on the final universe	Final asset universe; equal-weight vs. another fixed-weight scheme
2	Full-scale WP3-equivalent real-data backtest on 2016-2025 (both alphas, all models), including the 2022 stock-bond reversal	Whether to add a second regime-identifying variable (e.g. rates/VIX) given the taper-tantrum equity-blind-spot finding
3	Full-scale WP2-equivalent power/robustness study on the final universe and window; sensitivity to window/refit choices	Whether 2 vs. 3 regimes should be tested given real-data stress durations of 45-97 days
4	Draft results chapter; literature positioning writeup extended from output/literature_positioning.md; full reproducibility pass	Sign-off on final go/no-go framing and any portfolio-allocation extension (out of scope for the core forecasting question)

Generated by `scripts/generate_pilot_report.py` from `output/tables/wp2_summary.json`, `wp3_descriptive.json`, `wp3_walk_forward.json`, `wp3_stability.json`, and `wp4_reproducibility.json`. Machine: macOS-26.6.2-arm64-arm-64bit, 10 cores.